United States Patent	12409557
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Chou; Hung-Chun et al.

Training data generation device, training data generation method using the same and robot arm system using the same

Abstract

A training data generation device includes a virtual scene generation unit, an orthographic virtual camera, an object-occlusion determination unit, an object-occlusion determination unit and a perspective virtual camera. The virtual scene generation unit is configured for generating a virtual scene, wherein the virtual scene comprises a plurality of objects. The orthographic virtual camera is configured for capturing a vertical projection image of the virtual scene. The object-occlusion determination unit is configured for labeling an occluded-state of each object according to the vertical projection image. The perspective virtual camera is configured for capturing a perspective projection image of the virtual scene. The training data generation unit is configured for generating a training data of the virtual scene according to the perspective projection image and the occluded-state of each object.

Inventors: **Chou; Hung-Chun (Taipei, TW), Huang; Yu-Ru (Hualien, TW), Tsai; Dong-Chen (Miaoli, TW)**

Applicant: **INDUSTRIAL TECHNOLOGY RESEARCH INSTITUTE** (Hsinchu, TW)

Family ID: **85794485**

Assignee: **INDUSTRIAL TECHNOLOGY RESEARCH INSTITUTE (Hsinchu, TW)**

Appl. No.: **17/961921**

Filed: **October 07, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230150141 A1	May. 18, 2023

Foreign Application Priority Data

TW	110142173	Nov. 12, 2021

Publication Classification

Int. Cl.: **B25J11/00** (20060101); **B25J9/16** (20060101); **B25J18/00** (20060101); **B25J19/02** (20060101); **G06N20/00** (20190101)

U.S. Cl.:

CPC **B25J9/1697** (20130101); **B25J9/163** (20130101); **B25J9/1635** (20130101);

Field of Classification Search

CPC: B25J (9/1697); B25J (9/163); B25J (9/1635); B25J (11/00); B25J (18/00); B25J (19/021); G05B (2219/40053); G05B (2219/40532); G06N (20/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7313464	12/2006	Perreault	700/262	B25J 9/1666
9213908	12/2014	Benhimane et al.	N/A	N/A
10430663	12/2018	Chiang et al.	N/A	N/A
10442086	12/2018	Chen et al.	N/A	N/A
10713794	12/2019	He et al.	N/A	N/A
2005/0286767	12/2004	Hager et al.	N/A	N/A
2010/0111370	12/2009	Black et al.	N/A	N/A
2014/0039679	12/2013	Ando	N/A	N/A
2016/0136808	12/2015	Konolige	700/214	B25J 9/1612
2017/0206431	12/2016	Sun et al.	N/A	N/A
2017/0252922	12/2016	Levine et al.	N/A	N/A
2018/0250820	12/2017	Shimodaira et al.	N/A	N/A
2019/0355150	12/2018	Tremblay et al.	N/A	N/A
2019/0361672	12/2018	Odhner et al.	N/A	N/A
2019/0366545	12/2018	Mefford et al.	N/A	N/A
2020/0306960	12/2019	Handa et al.	N/A	N/A
2020/0342652	12/2019	Rowell	N/A	G06V 10/82
2020/0361083	12/2019	Mousavian et al.	N/A	N/A
2020/0363815	12/2019	Mousavian et al.	N/A	N/A
2021/0081752	12/2020	Chao et al.	N/A	N/A
2021/0331311	12/2020	Fu	N/A	G06V 10/764
2022/0288783	12/2021	Sundermeyer	N/A	G06T 7/55
2022/0301304	12/2021	Hampali	N/A	G06T 7/73
2022/0405506	12/2021	Taamazyan	N/A	G06T 7/50

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111666919	12/2019	CN	N/A
110555878	12/2020	CN	N/A
112770875	12/2020	CN	N/A

201738056	12/2016	TW	N/A
I708041	12/2019	TW	N/A
I708190	12/2019	TW	N/A
WO2018/035216	12/2017	WO	N/A
WO2020/106908	12/2019	WO	N/A
WO 2020/132372	12/2019	WO	N/A
WO 2020/160394	12/2019	WO	N/A
WO 2020/180863	12/2019	WO	N/A
WO 2020/185334	12/2019	WO	N/A
WO 2021/025800	12/2020	WO	N/A

OTHER PUBLICATIONS

Baimukashev et al., “Deep Learning Based Object Recognition Using Physically-Realistic Synthetic Depth Scenes”, Machine Learning and Knowledge Extraction, vol. 1, 2019, pp. 883-903. cited by applicant
Ehsani et al., “SeGAN: Segmenting and Generating the Invisible”, IEEE Xplore, 2018, pp. 6144-6153. cited by applicant
Ke et al., “Deep Occlusion-Aware Instance Segmentation with Overlapping BiLayers”, IEEE Xplore, 2021, pp. 4019-4028. cited by applicant
Wada et al., “Instance Segmentation of Visible and Occluded Regions for Finding and Picking Target from a Pile of Objects”, IEEE/RSJ International Conference on Intelligent Robots and Systems, 2018, pp. 2048-2055. cited by applicant
Chinese Office Action and Search Report for Chinese Application No. 202111536541.2, dated Mar. 10, 2025. cited by applicant

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Primary Examiner: Mott; Adam R

Assistant Examiner: Keniry; Heather J

Attorney, Agent or Firm: Birch, Stewart, Kolasch & Birch, LLP

Background/Summary

(1) This application claims the benefit of Taiwan application Serial No. 110142173, filed Nov. 12, 2021, the subject matter of which is incorporated herein by reference.
TECHNICAL FIELD
(2) The disclosure relates in general to a training data generation device, a training data generation method using the same and a robot arm system using the same.
BACKGROUND
(3) In the field of picking-up object, when a robotic arm picks up an actual object that is occluded (covered by another object), it is easy to cause problems of, for example, object jamming, object damage and/or failure to pick up the object. Therefore, how to improve the success rate of picking up the object is one of the goals of those skilled in the art.
SUMMARY
(4) According to an embodiment, a training data generation device is provided. The training data generation device includes a virtual scene generation unit of a processor, an orthographic virtual camera of the processor, an object-occlusion determination unit of the processor, a perspective virtual camera of the processor and a training data generation unit of the processor. The virtual

scene generation unit is configured for generating a virtual scene, wherein the virtual scene comprises a plurality of objects. The orthographic virtual camera is configured for capturing a vertical projection image of the virtual scene. The object-occlusion determination unit is configured for labeling an occluded-state of each object according to the vertical projection image. The perspective virtual camera is configured for capturing a perspective projection image of the virtual scene. The training data generation unit is configured for generating a training data of the virtual scene according to the perspective projection image and the occluded-state of each object.
(5) According to another embodiment, a training data generation method is provided. The training data generation method implemented by a processor includes the following steps: generating a virtual scene, wherein the virtual scene comprises a plurality of objects; capturing a vertical projection image of the virtual scene; labeling an occluded-state of each object according to the vertical projection image; capturing a perspective projection image of the virtual scene; and generating training data of the virtual scene according to the perspective projection image and the occluded-state of each object.
(6) According to another embodiment, a robotic arm system is provided. The robotic arm system includes a learning model generated by a training data generation method. The training data generation method implemented by a processor includes the following steps: generating a virtual scene, wherein the virtual scene comprises a plurality of objects; capturing a vertical projection image of the virtual scene; labeling an occluded-state of each object according to the vertical projection image; capturing a perspective projection image of the virtual scene; and generating training data of the virtual scene according to the perspective projection image and the occluded-state of each object. The robotic arm system includes a robotic arm, a perspective camera and a controller. The perspective camera is disposed on the robotic arm and configured for capturing an actual object image of a plurality of actual objects. The controller is configured for analyzing the actual object image according to the learning model, accordingly determining an actual occluded-state of each actual object, and controlling the robotic arm to pick up the actual object whose the actual occluded-state is un-occluded.
(7) The above and other aspects of the disclosure will become better understood with regard to the following detailed description of the preferred but non-limiting embodiment (s). The following description is made with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A illustrates a functional block diagram of a training data generation device according to an embodiment of the present disclosure;
(2) FIG. **1**B illustrates a schematic diagram of a robotic arm system according to an embodiment of the present disclosure;
(3) FIG. **2** illustrates a schematic diagram of a virtual scene of FIG. **1**;
(4) FIG. **3**A illustrates a schematic diagram of a perspective virtual camera of FIG. **1** capturing a perspective projection image;
(5) FIG. **3**B illustrates a schematic diagram of the perspective projection image of the virtual scene of FIG. **2**;
(6) FIG. **4**A illustrates a schematic diagram of an orthographic virtual camera of FIG. **1** capturing a vertical projection image;
(7) FIG. **4**B illustrates a schematic diagram of the vertical projection image of the virtual scene of FIG. **2**;
(8) FIG. **5**A illustrates a schematic diagram of a target object scene of the object of FIG. **2** being a target object;

(9) FIG. **5**B illustrates a schematic diagram of a vertical projection object image of the target object scene of FIG. **5**A;
(10) FIG. **5**C illustrates a schematic diagram of the vertical projection object image of FIG. **4**B;
(11) FIG. **6** illustrates a flowchart of a training data generation method using the training data generation device of FIG. **1**A; and
(12) FIG. **7** illustrates a flowchart of the method of determining the occluded-state in step S**130** of FIG. **6**.
(13) In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments could be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.
DETAILED DESCRIPTION
(14) Referring to FIGS. **1**A to **4**B, FIG. **1**A illustrates a functional block diagram of a training data generation device **100** according to an embodiment of the present disclosure, FIG. **1**B illustrates a schematic diagram of a robotic arm system **1** according to an embodiment of the present disclosure, FIG. **2** illustrates a schematic diagram of a virtual scene **10** of FIG. **1**, FIG. **3**A illustrates a schematic diagram of a perspective virtual camera **140** of FIG. **1** capturing a perspective projection image **10***e*, FIG. **3**B illustrates a schematic diagram of the perspective projection image **10***e* of the virtual scene **10** of FIG. **2**, FIG. **4**A illustrates a schematic diagram of an orthographic virtual camera **120** of FIG. **1** capturing a vertical projection image **10***v*, and FIG. **4**B illustrates a schematic diagram of the vertical projection image **10***v* of the virtual scene **10** of FIG. **2**.
(15) As shown in FIG. **1**A, the training data generation device **100** includes a virtual scene generation unit **110**, the orthographic virtual camera **120**, an object-occlusion determination unit **130**, the perspective virtual camera **140** and a training data generation unit **150**. At least one of the virtual scene generation unit **110**, the orthographic virtual camera **120**, the object-occlusion determination unit **130**, the perspective virtual camera **140** and the training data generation unit **150** is, for example, software, firmware or a physical circuit formed by semiconductor processes. At least one of the virtual scene generation unit **110**, the orthographic virtual camera **120**, the object-occlusion determination unit **130**, the perspective virtual camera **140** and the training data generation unit **150** could be integrated into a single unit or integrated into a processor or a controller.
(16) As shown in FIG. **1**A, the virtual scene generation unit **110** is configured to generate the virtual scene **10** including a number of objects **11** to **14**. The orthographic virtual camera **120** is configured for capturing the vertical projection image **10***v* of the virtual scene **10** (the vertical projection image **10***v* is shown in FIG. **4**B). The object-occlusion determination unit **130** is configured to label an occluded-state ST of each of the objects **11** to **14** (the objects **11** to **14** are shown in FIG. **2**) according to the vertical projection image **10***v*. The perspective virtual camera **140** is configured to capture the perspective projection image **10***e* of the virtual scene **10**. The training data generation unit **150** is configured for generating a training data TR of the virtual scene **10** according to the perspective projection image **10***e* and the occluded-state ST of each object **10**. For example, each training data TR includes the perspective projection image **10***e* and the occluded-state ST of each object **10**. Compared with the perspective projection image **10***e*, the vertical projection image **10***v* could much better reflect the actual occluded-state of the objects **11** to **14**, so that the occluded-state ST determined according to the vertical projection image **10***v* could much better match the actual occluded-state of the object. As a result, a learning model M1 (the learning model M1 is shown in FIG. **1**A) which is subsequently generated according to the training data TR could be used to more accurately determine the actual occluded-state of an actual object. The aforementioned occluded-state ST includes, for example, a number of categories, for example, two categories such as “occluded” and “un-occluded”.

(17) The so-called “virtual camera” herein, for example, projects a three-dimensional (3D) scene onto a reference surface (for example, a plane) to become a two-dimensional (2D) image by using computer vision technology. In addition, the aforementioned virtual scene **10**, the perspective projection image **10***e* and the vertical projection image **10***v* could be data during the operation of the unit, not necessarily real image.
(18) As shown in FIG. **1**A, the training data generation device **100** could generate a number of the virtual scenes **10** and accordingly generate the training data TR of each virtual scene **10**. A machine learning unit **160** could generate the learning model M1 by learning the occluding determination of the object according to a number of the training data TR. The machine learning unit **160** could be a sub-element of the training data generation device **100**, or could be independently disposed with respect to the training data generation device **100**, or the machine learning unit **160** could be disposed on the robotic arm system **1** (the robotic arm system **1** is shown in FIG. **1**B). In addition, the learning model M1 generated by the machine learning unit **160** is obtained by using, for example, Mask R-CNN (object segmentation), Faster-R-CNN (object detection) or other types of machine learning techniques or algorithms.
(19) As shown in FIG. **1**B, the robotic arm system **1** includes a robotic arm **1**A, a controller **1**B and a perspective camera **1**C. The perspective camera **1**C is an actual (physical) camera. The robotic arm **1**A could capture (or pick up) an actual object **22**, for example, by gripping or sucking (vacuum suction or magnetic suction). In the actual object picking application, the perspective camera **1**C could capture the actual object images P1 of the underneath actual objects **22**. The controller **1**B is electrically connected to the robotic arm **1**A and the perspective camera **1**C and could determine the actual occluded-state of the actual object **22** according to (or by analyzing) the actual object image P1 and control the robotic arm **1**A to capture the underneath actual object **22**. The controller **1**B could load in the aforementioned trained learning model M1. Since the learning model M1 (the learning model M1 is shown in FIG. **1**A) trained and generated according to the training data TR could be used to more accurately determine the actual occluded-state of the actual object **22**, the controller **1**B of the robotic arm system **1** could accurately determine the actual occluded-state of the actual object **22** in the actual scene according to the learning model M1 and then control the robot arm **1**A to successfully capture (or pick up) the actual object **22**. For example, the robot arm **1**A captures the actual object **22** that is not occluded at all (for example, the actual object **22** whose actual occluded-state is “un-occluded”). As a result, in the actual object picking application, it could increase the success rate of picking up objects and/or improve the problems of object damage and object jamming.
(20) In addition, as shown in FIG. **1**B, the perspective camera **1**C disposed on the robotic arm system **1** captures the actual object image P1 by the perspective projection principle or method. The actual object image P1 captured by the perspective camera **1**C and the perspective projection image **10***e* based on the learning model M1 are obtained by the same perspective projection principle or method, and thus the controller **1**B could accurately determine the occluded-state of the object.
(21) As shown in FIGS. **1**A and **2**, the virtual scene generation unit **110** generates the virtual scene **10** by using, for example, V-REP (Virtual Robot Experiment Platform) technology. Each virtual scene **10** is, for example, a randomly generated 3D scene. As shown in FIG. **2**, the virtual scene **10** includes, for example, a virtual container **15** and a number of virtual objects **11** located within the virtual container **15**. The stacking status of the objects, the arrangement status of the objects and/or the number of objects in each virtual scene **10** are also randomly generated by the virtual scene generation unit **110**. For example, the objects are disorderly stacked, randomly arranged and/or closely arranged. Therefore, the stacking status of the objects, the arrangement status of the objects and/or the number of the objects in each virtual scene **10** are not completely identical. In addition, the objects in the virtual scene **10** could be any type of objects, for example, hand tools, stationery, plastic bottle, food materials, mechanical parts and other objects used in factories, such as a machinery processing factory, a food processing factory, a stationery factory, a resource recycling

factory, an electronic device processing factory, an electronic device assembly factory, etc. The 3D model of the object and/or the container **15** could be made in advance by using modeling software, and the virtual scene generation unit **110** generates the virtual scene **10** according to the pre-modeled 3D model of the object and/or the container **15**.

(22) As shown in FIG. **3**A, the perspective virtual camera **140**, as like general perspective camera (with a conical angle-of-view A1), photograph the virtual scene **10** to obtain the perspective projection image **10***e* (the perspective projection image **10***e* is shown in FIG. **3**B). As shown in FIG. **3**B, the perspective projection image **10***e* is, for example, a perspective projection 2D image of the virtual scene **10** (3D scene), wherein the perspective projection 2D image includes a number of perspective projection object images **11***e* to **14***e* corresponding to the objects **11** to **14** in FIG. **2**. Viewed from the perspective projection image **10***e*, the perspective projection object images **11***e* to **13***e* overlap each other. For example, the perspective projection object image **12***e* is occluded (or shielded, or covered) by the perspective projection object image **11***e*. Thus, the object occluded-state of the perspective projection object image **12***e* is determined to be "occluded" accordion to the perspective projection image **10***e*, but it is not the actual occluded-state of the perspective projection object image **12***e* in the virtual scene **10**.

(23) As shown in FIG. **4**A, the orthographic virtual camera **120** photograph, with a vertical line-of-sight V1, the virtual scene **10** to obtain the vertical projection image **10***v* (the vertical projection image **10***v* is shown in FIG. **4**B). As shown in FIG. **4**B, the vertical projection image **10***v* is, for example, a vertical projection 2D image of the virtual scene **10** (3D scene), wherein the vertical projection 2D image includes a number of vertical projection object images **11***v* to **14***v* corresponding to the objects **11** to **14** of FIG. **2**. As viewed from the vertical projection image **10***v*, the vertical projection object image **12***v* is not occluded by the vertical projection object image **11***v*. Thus, the actual occluded-state of the vertical projection object image **12***v* is determined to be "un-occluded" according to the vertical projection image **10***v*. Due to the occluded-state ST of the object in the embodiment of the present disclosure being determined according to the vertical projection image **10***v*, it could reflect the actual occluded-state of each object in the virtual scene **10**.

(24) As shown in FIG. **4**A, the aforementioned objects **11** to **14** (or the container **15**) are placed relative to a carrying surface G1, and the vertical projection image **10***v* of FIG. **4**B is generated by viewing the virtual scene **10** in the vertical line-of-sight V1. The vertical line-of-sight V1 referred to herein is, for example, perpendicular to the carrying surface G1. In the present embodiment, the carrying surface G1 is, for example, a horizontal surface. In another embodiment, the carrying surface G1 could be an inclined surface, and the vertical line-of-sight V1 is also perpendicular to the inclined carrying surface G1. Alternatively, the direction of the vertical line-of-sight V1 could be defined to be consistent with or parallel to a direction of the robotic arm (not shown) picking up the object.

(25) In addition, the embodiments of the present disclosure could determine the occluded-state of the object using a variety of methods, and one of which will be described below.

(26) Referring to FIGS. **5**A to **5**C, FIG. **5**A illustrates a schematic diagram of a target object scene **12**′ of the object **12** of FIG. **2** being a target object, FIG. **5**B illustrates a schematic diagram of a vertical projection object image **12***v*′ of the target object scene **12**′ of FIG. **5**A, and FIG. **5**C illustrates a schematic diagram of the vertical projection object image **12***v* of FIG. **4**B.

(27) The object-occlusion determination unit **130** is further configured for hiding the objects other than the target one in the virtual scene, and correspondingly generating the target object scene of the target one. The orthographic virtual camera **120** is further configured for obtaining the vertical projection object image of the target object scene. The object-occlusion determination unit **130** is further configured for obtaining a difference value between the vertical projection object image of the target object scene and the vertical projection object image of the target object of the vertical projection image and determining the occluded-state ST according to the difference value. The

object-occlusion determination unit **130** is configured for determining whether the difference value is smaller than a preset value, determining that the occluded-state of the target object is "un-occluded" when the difference value is smaller than the preset value, and determining the occluded-state of the target object is "occluded" when the difference value is not smaller than the preset value. The unit of the aforementioned "preset value" is, for example, the number of pixels. The embodiments of the present disclosure do not limit the numerical range of the "preset value", and it could depend on actual conditions.

(28) In the case of the object **12** being the target object, as shown in FIG. **5**A, the object-occlusion determination unit **130** hides the objects **11** and **13**-**14** other than the object **12** in the virtual scene **10** of FIG. **2**. For example, the object-occlusion determination unit **130** sets the object **12** as "visible state" while sets the other objects **11** and **13** to **14** as "invisible state". Then, the object-occlusion determination unit **130** accordingly generates the target object scene **12**′ of the object **12**. As shown in FIG. **5**B, the orthographic virtual camera **120** captures the vertical projection object image **12***v*′ of the target object scene **12**′. The object-occlusion determination unit **130** obtains the difference value between the vertical projection object image **12***v*′ of the target object scene **12**′ and the vertical projection object image **12***v* (vertical projection object image **12***v* is shown in FIG. **5**C) of the object **12** of the vertical projection image **10***v*, and determines the occluded-state ST according to the difference value. If the difference between the vertical projection object image **12***v* and the vertical projection object image **12***v*′ is substantially equal to 0 or less than a preset value, it means that the object **12** is not occluded, and accordingly the object-occlusion determination unit **130** could determine that the occluded-state ST of the object **12** is "un-occluded". Conversely, if the difference between the vertical projection object image **12***v* and the vertical projection object image **12***v*′ is not equal to 0 or not less than the preset value, it means that the object **12** is occluded, and accordingly the object-occlusion determination unit **130** could determine that the occluded-state ST of the object **12**. ST is "occluded". In addition, the aforementioned preset value is greater than 0, for example, 50, but it could also be any integer less than or greater than 50.

(29) The occluded-state of each object in the virtual scene **10** could be determined by using the same manner. For example, the training data generation device **100** could determine, by using the same manner, the occluded-states ST of the objects **11**, **13** and **14** in the virtual scene **10** of FIG. **2** as "occluded", "un-occluded" and "un-occluded" respectively, as shown in FIG. **4**B.

(30) Referring to FIG. **6**, FIG. **6** illustrates a flowchart of a training data generation method using the training data generation device **100** of FIG. **1**A.

(31) In step S**110**, as shown in FIG. **2**, the virtual scene generation unit **110** generates the virtual scene **10** by using, for example, the V-REP technology. The virtual scene **10** includes a number of the virtual objects **11** to **14** and the virtual container **15**.

(32) In step S**120**, as shown in FIG. **4**B, the orthographic virtual camera **120** captures the vertical projection image **10***v* of the virtual scene **10** by using computer vision technology. The capturing method is, for example, to convert the 3D virtual scene **10** into the 2D vertical projection image **10***v* by using computer vision technology. However, as long as the 3D virtual scene **10** could be converted into the 2D vertical projection image **10***v*, the embodiment of the present disclosure does not limit the processing technology, and it could even be a suitable conventional technology.

(33) In step S**130**, as shown in FIG. **1**B, the object-occlusion determination unit **130** labels the occluded-state ST of each of the object **11** to **14** according to the vertical projection image **10***v*. The method of determining the occluded-state ST of the object has been described above, and the similarities will not be repeated here.

(34) In step S**140**, as shown in FIG. **3**B, the perspective virtual camera **140** captures the perspective projection image **10***e* of the virtual scene **10** by using, for example, computer vision technology. The capturing method is, for example, to convert the 3D virtual scene **10** into the 2D perspective projection image **10***e* by using computer vision technology. However, as long as the 3D virtual scene **10** could be converted into the 2D perspective projection image **10***e*, the embodiment of the

present disclosure does not limit the processing technology, and it could even be a suitable conventional technology.

(35) In step S**150**, as shown in FIG. **1**B, the training data generation unit **150** generates the training data TR of the virtual scene **10** according to the perspective projection image **10***e* and the occluded-state ST of each of the objects **11** to **14**. For example, the training data generation unit **150** integrates the data of the perspective projection image **10***e* and the data of the occluded-states ST of the objects **11** to **14** into one training data TR.

(36) After obtaining one training data TR, the training data generation device **100** could generate the training data TR of the next virtual scene **10** by repeating steps S**110** to S**150**, wherein such training data TR includes the perspective projection image **10***e* and the occluded-state ST of each object of the next virtual scene **10**. According to such principle, the training data generation device **100** could obtain N training data TR. The embodiments of the present disclosure do not limit the value of N, and the value could be a positive integer equal to or greater than 2, such as 10, 100, 1000, 10000, or even more or less.

(37) Referring to FIG. **7**, FIG. **7** illustrates a flowchart of the method of determining the occluded-state in step **130** of FIG. **6**. The following description takes the object **12** of FIG. **2** as the target object, and the determining method for the occluded-state ST of the other objects is the same, and the similarities will not be repeated here.

(38) In step S**131**, the object-occlusion determination unit **130** hides the objects **11** and **13** to **14** other than the object **12** in the virtual scene **10**, and accordingly generates the target object scene **12**′ of the object **12**, as shown in FIG. **5**A.

(39) In step S**132**, the orthographic virtual camera **120** captures the vertical projection object image **12***v*′ of the target object scene **12**′ by using, for example, computer vision technology, as shown in FIG. **5**B. The capturing method of the vertical projection object image **12***v*′ is the same as that of the aforementioned vertical projection image **10***v*.

(40) In step S**133**, the object-occlusion determination unit **130** obtains the difference value between the vertical projection object image **12***v*′ (the vertical projection object image **12***v*′ is shown in FIG. **5**B) and the vertical projection object image **12***v* (the vertical projection object image **12***v* is shown in FIG. **5**C) of the vertical projection image **10***v* (the vertical projection image **10***v* is shown in FIG. **4**B). For example, the object-occlusion determination unit **130** performs a subtraction calculation between the vertical projection object image **12***v*′ and the vertical projection object image **12***v*, and uses the absolute value of result of the subtraction calculation as the difference value.

(41) In step S**134**, the object-occlusion determination unit **130** determines whether the difference value is smaller than the preset value. If the difference value is less than the preset value, the process proceeds to step S**135**, and the object-occlusion determination unit **130** determines that the occluded-state ST of the object **12** is “un-occluded”; if the difference value is not less than the preset value, the process proceeds to step S**136**, and the object-occlusion determination unit **130** determines that the occluded-state ST of the object **12** is “occluded”.

(42) In an experiment, the learning model M1 is obtained, according to 2000 virtual scenes, by using mask-r-cnn of machine learning technology. The controller **1**C performs, for a number of the actual object images of a number of the actual scenes (the decoration of the actual objects in each scene may be different), the determination of the actual occluded-state of the actual object according to the learning model M1. According to the experimental results, a mean average precision (mAP) obtained by using the training data generation method of the embodiment of the present disclosure ranges between 70% and 95%, or even higher, while the mean average precision obtained by using the conventional training data generation method is normally not more than 50%. The aforementioned mean average precision is obtained under 0.5 IoU (Intersection Over Union). In comparison with the conventional training data generation method, the training data generation method of the disclosed embodiment does have much higher mean average precision.

(43) To sum up, an embodiment of the present disclosure provides a training data generation device

and a training data generation method using the same, which could generate at least one training data. The training data includes the occluded-state of each object in the virtual scene and the perspective projection image of the virtual scene. The occluded-state of each object is obtained according to the vertical projection image of the virtual scene captured by the orthographic virtual camera, and thus it could reflect the actual occluded-state of the object. The perspective projection image is obtained by using the projection principle which is the same as that of the general perspective camera for actual picking-up object, and thus the mechanical system could accurately determine the occluded-state of the actual object. As a result, the learning model generated according to (or learning) the aforementioned training data could improve the mean average precision of the robotic arm system in actual picking-up object.
(44) It will be apparent to those skilled in the art that various modifications and variations could be made to the disclosed embodiments. It is intended that the specification and examples be considered as exemplary only, with a true scope of the disclosure being indicated by the following claims and their equivalents.

Claims

1. A training data generation device, comprising: a virtual scene generation unit of a processor configured for generating a virtual scene, wherein the virtual scene comprises a plurality of objects; an orthographic virtual camera of the processor configured for capturing a vertical projection image of the virtual scene; an object-occlusion determination unit of the processor configured for labeling an occluded-state of each object according to the vertical projection image; a perspective virtual camera of the processor configured for capturing a perspective projection image of the virtual scene; and a training data generation unit of the processor configured for generating a training data of the virtual scene according to the perspective projection image and the occluded-state of each object.
2. The training data generation device as claimed in claim 1, wherein the object-occlusion determination unit is further configured for hiding the objects other than a target object in the virtual scene and correspondingly generating a target object scene of the target object; the orthographic virtual camera is further configured for obtaining a vertical projection object image of the target object scene; wherein the object-occlusion determination unit is further configured for obtaining a difference value between the vertical projection object image of the target object scene and a vertical projection object image of the target object of the vertical projection image and determining the occluded-state according to the difference value.
3. The training data generation device as claimed in claim 2, wherein the object-occlusion determination unit is further configured for determining whether the difference value is smaller than a preset value, determining that the occluded-state of the target object is un-occluded if the difference value is smaller than the preset value and determining that the occluded-state of the target object is occluded if the difference value is not smaller than the preset value.
4. The training data generation device as claimed in claim 1, wherein the objects are placed relative to a carrying surface, the perspective virtual camera is further configured for capturing the perspective projection image in a vertical line-of-sight, and the vertical line-of-sight is perpendicular to the carrying surface.
5. The training data generation device as claimed in claim 1, wherein the virtual scene comprises a container within which the objects are placed, the container is placed relative to a carrying surface, the vertical projection image is generated by viewing the virtual scene in the vertical line-of-sight, and the vertical line-of-sight is perpendicular to the carrying surface.
6. The training data generation device as claimed in claim 1, wherein the occluded-state comprises a state which is occluded or un-occluded.
7. A training data generation method implemented by a processor, comprising: generating a virtual

scene, wherein the virtual scene comprises a plurality of objects; capturing a vertical projection image of the virtual scene; labeling an occluded-state of each object according to the vertical projection image; capturing a perspective projection image of the virtual scene; and generating a training data of the virtual scene according to the perspective projection image and the occluded-state of each object.

8. The training data generation method as claimed in claim 7, wherein the step of labeling the occluded-state of each object according to the vertical projection image comprises: labeling the occluded-state of each object to be occluded or un-occluded according to the vertical projection image.

9. The training data generation method as claimed in claim 7, further comprising: hiding the objects other than a target object in the virtual scene; correspondingly generating a target object scene of the target object; obtaining a vertical projection object image of the target object scene; obtaining a difference value between the vertical projection object image of the target object scene and a vertical projection object image of the target object of the vertical projection image; and determining the occluded-state according to the difference value.

10. The training data generation method as claimed in claim 9, further comprising: determining whether the difference value is smaller than a preset value; determining that the occluded-state of the target object is un-occluded if the difference value is smaller than the preset value; and determining that the occluded-state of the target object is occluded if the difference value is not smaller than the preset value.

11. The training data generation method as claimed in claim 7, wherein in the step of generating the virtual scene, the objects are placed relative to a carrying surface; the training data generation method further comprises: capturing the perspective projection image in a vertical line-of-sight, wherein the vertical line-of-sight is perpendicular to the carrying surface.

12. The training data generation method as claimed in claim 7, wherein in the step of generating the virtual scene, the virtual scene comprises a container within which the objects are placed, and the container is placed relative to a carrying surface; the training data generation method further comprises: capturing the vertical projection image in a vertical line-of-sight, wherein the vertical line-of-sight is perpendicular to the carrying surface.

13. A robotic arm system comprising a learning model generated by the training data generation method according to claim 7, comprising: a robotic arm; a perspective camera disposed on the robotic arm and configured for capturing an actual object image of a plurality of actual objects; a controller configured for analyzing the actual object image according to the learning model, accordingly determining an actual occluded-state of each actual object, and controlling the robotic arm to pick up the actual object whose the actual occluded-state is un-occluded.